

DirSource

The `DirSource()` function accepts a directory source that contains files as input for further processing. It accepts the following arguments:

Argument	Description
directory	A character vector of file path names
encoding	Character encoding
pattern	Optional regular expression to match files
recursive	Logical value to recurse into directories
ignore.case	Logical value for case-sensitive pattern matching
mode	Character strings - "binary", "text", or "" (No read)

The following example demonstrates the use of the `DirSource()` function.

```
> d <- DirSource(system.file("texts", "txt", package="tm"))
> d
$encoding
[1] ""

$length
[1] 5

$position
[1] 0

$reader
function (elem, language, id)
{
  if (!is.null(elem$uri))
    id <- basename(elem$uri)
  PlainTextDocument(elem$content, id = id, language =
language)
}
<bytecode: 0x555cec9d12c8>
<environment: namespace:tm>

$mode
[1] "text"
$filelist
[1] "/home/guest/R/x86_64-pc-linux-gnu-library/4.1/tm/texts/
txt/ovid_1.txt"
...
attr(,"class")
[1] "DirSource"      "SimpleSource" "Source"
VectorSource
```

The Wikipedia page on 'Open Source' at https://en.wikipedia.org/wiki/Open_source has been downloaded as a PDF, converted to plain text using the Ghostscript (gs) tool and saved as 'Open_source.txt'. We can read the content from this file into various lines, and create a vector source of input using the `VectorSource()` function. This data can be kept in memory for processing the R object using the `VCorpus()` function, as demonstrated below:

```
> input <- "Open_source.txt"
> connection <- file(input)
> lines <- readLines(connection)

> lines

[1] "Open source"

[2] "Open source is source code that is made freely
available for possible modification and redistribution."
[3] "Products include permission to use the source
code,[1] design documents,[2] or content of the product. The"
[4] "open-source model is a decentralized software
development model that encourages open"
[5] "collaboration.[3][4] A main principle of open-source
software development is peer production, with products"
...
> feed <- VCorpus(VectorSource(lines))
> feed
<<VCorpus>>
Metadata: corpus specific: 0, document level (indexed): 0
Content: documents: 1465

> summary(feed)
      Length Class      Mode
1      2      PlainTextDocument list
2      2      PlainTextDocument list
...
```

The `VCorpus()` function accepts the following arguments:

Argument	Description
x	A source object for VCorpus
readerControl	A list of control parameters; reader function for reading x; default language is 'en'

stemCompletion

The `stemCompletion()` function provides heuristic completion of words. For the input feed 'Open_source.txt' file, the 'op' and 'so' stem constructs complete to 'open-source' and 'software', as shown below: